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Mobility Lens Project Report

Emulator and API Level

I ran the app on a **Pixel 9** emulator running **API 37.1**.

SDK Configuration

- **minSdk: 26**
 - sets the lowest Android version the app can install on. I picked minSdk 26 because it covers about 98% of active devices, and it skips the older pre-Oreo background/notification behavior that works differently from modern Android. My app doesn't use background tasks or notifications anyway, so this didn't cost me anything.
- **targetSdk: 37**
 - tells the system which Android version I built the app for, so it gets the latest behavior instead of older compatibility fallbacks.
- **compileSdk: 37**
 - what SDK version I'm compiling against, it doesn't affect what devices can run the app, only what APIs I can use while building.

Project Structure

MainActivity.kt (app/src/main/java/com/jimmyh/mobilitylens/) is where the app starts. It has onCreate(), the MobilityDimension data class (holds each dimension's name/description/implication), and the main composable with all the UI and state.

AndroidManifest.xml (app/src/main/) tells Android about the app, sets MainActivity as the launcher screen, sets the app name/icon, and lists permissions.

The app-level Gradle file (app/build.gradle.kts) sets up how the app module builds: SDK versions, Compose settings, and dependencies.

The version catalog (libs.versions.toml) keeps all dependency versions in one file instead of scattering version numbers everywhere.

strings.xml (app/src/main/res/values/) holds all the text the user sees instead of hardcoding it in Kotlin, keeps things organized and easier to change later.

Rotation Observation

I went to a dimension that wasn't the first one, typed something in the text field, hit Check to see the message, then rotated the emulator. After rotating, the app reset back to the first dimension, the text field was empty, and the message was gone.

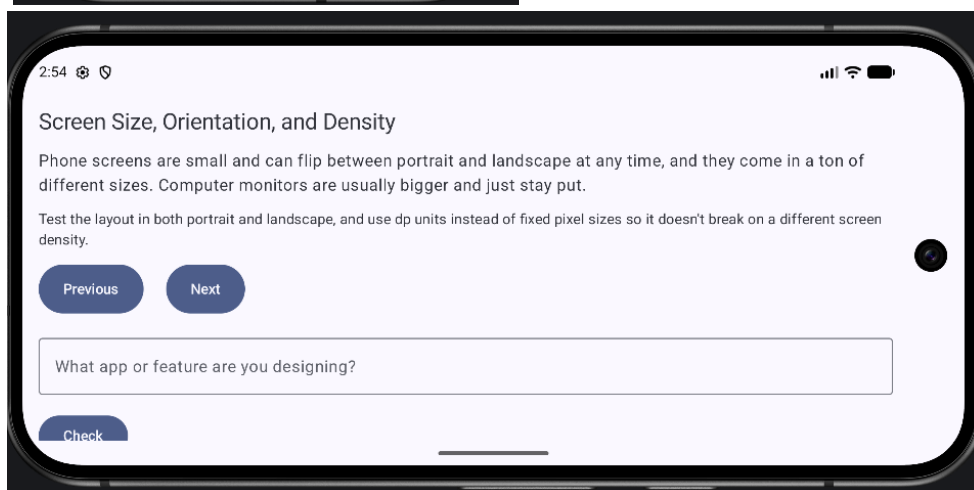
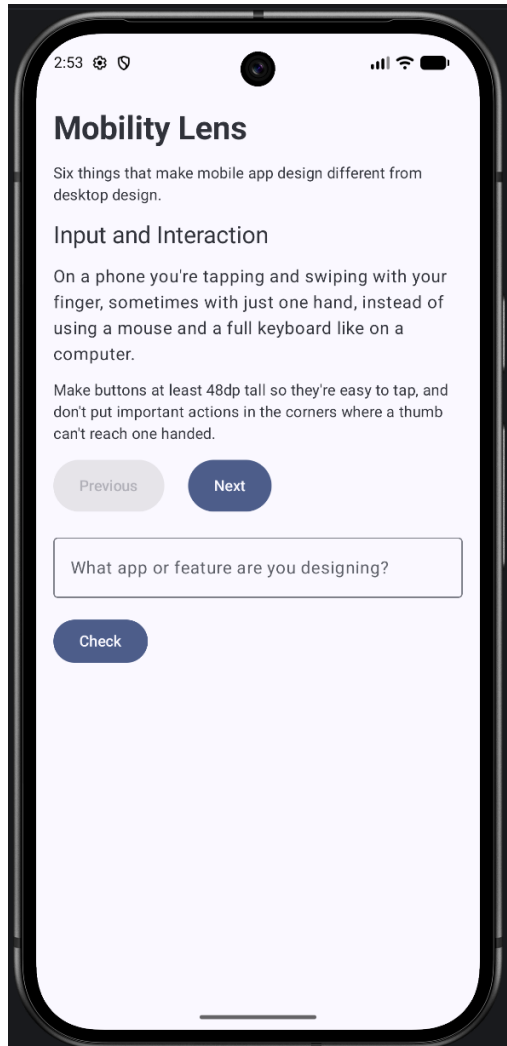
This happened because I used `remember` to hold my state. `remember` only survives recomposition, not the Activity actually getting destroyed and rebuilt, which is what happens on rotation. So all my state just went back to its starting values.

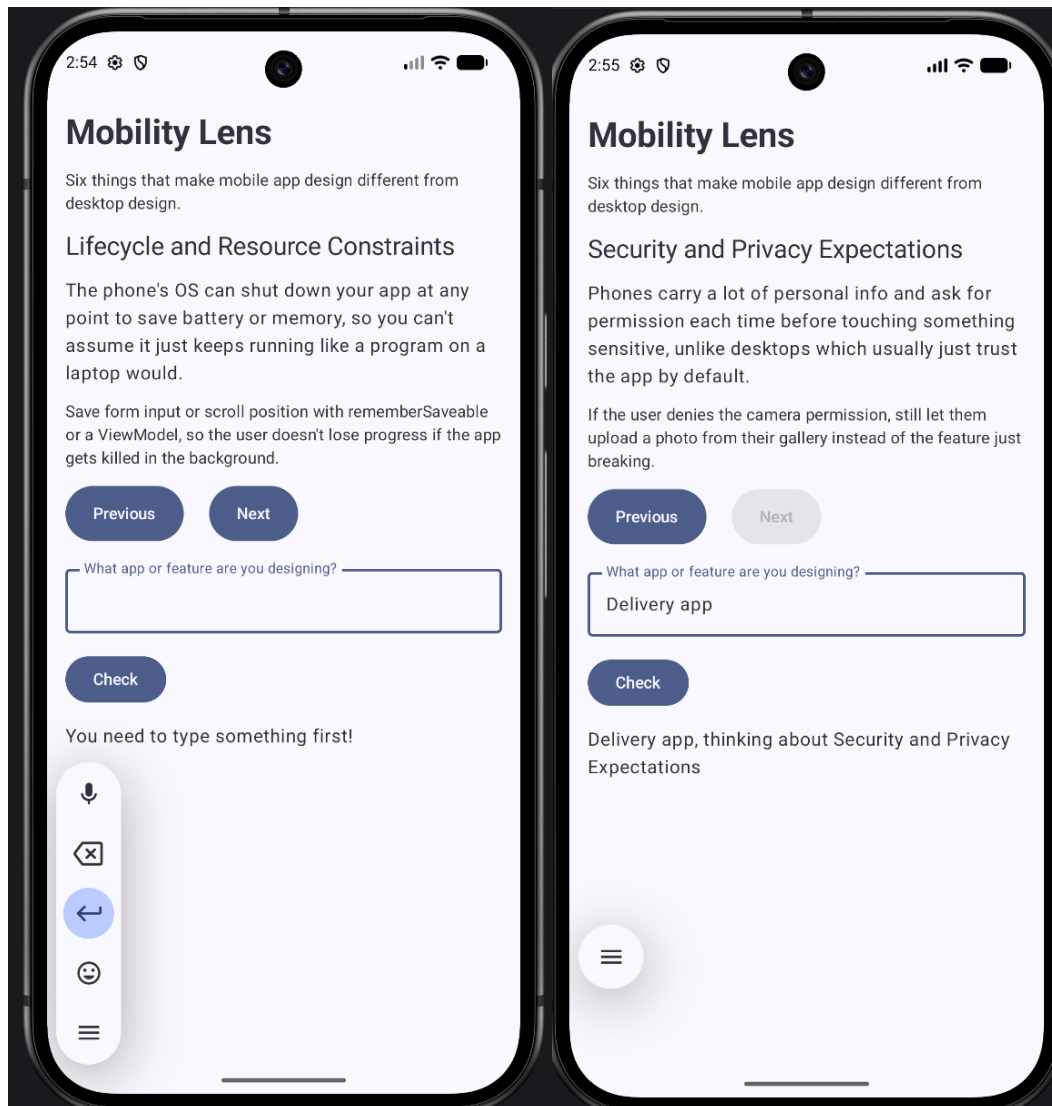
Apps need to hold onto this kind of state because rotating (or other config changes) happens all the time on phones, way more than on desktop. If the app just wipes out whatever the user typed every time they rotate their phone, that wouldn't be a good experience for the users.

Mobile Design Decision

I made the Previous/Next buttons big with a decent amount of padding instead of small tight buttons. On desktop you click with a mouse, which is precise. On a phone you're tapping with a finger, which isn't nearly as precise, so buttons need to be bigger, so people don't miss the taps. This ties directly into the "Input and Interaction" dimension description.

Screenshots





Generative-AI disclosure

Tool used: Claude Sonnet 5 chat

Prompts: I asked how to structure the app, why the screen wasn't scrollable when rotated sideways, how Button's onClick logic works, and for typography suggestions for different sections of text. I also asked it to fix grammar and sentence structure in my strings.xml text.

Suggestions produced: It explained landscape mode was cutting off content because the Column had no scroll modifier and suggested adding `.verticalScroll(rememberScrollState())`, and suggested distinct text styles in Type.kt so each text section looked visually different.

What I accepted, changed, or rejected: I accepted scroll fix and the typography suggestions but picked the specific font weights/sizes myself. Paraphrased strings.xml text but reworded some lines afterward.

How I verified the code: I ran the app in the emulator after each change, tested all six dimensions, the text input, the blank-input warning, a valid Check response, and rotated the emulator to confirm the scroll.